MATH 101 Unit 3: Polynomial, Rational, and Radical Functions

Practice 2 | Version 202610122020.41 | 30 minutes

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right. Basic four-function calculator only.

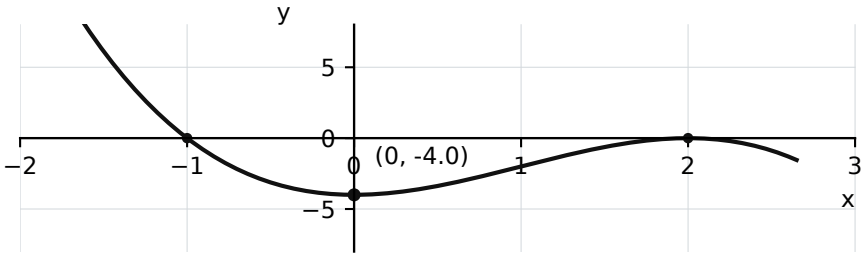
QUESTIONS	ANSWERS
1. Find (a) the vertex and the minimum or maximum value (state which); (b) all zeros. $f(x) = 2x^2 - 18$	1. (a) Vertex 6 pts (a) Minimum 4 pts (b) Zeros 10 pts
2. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts. $f(x) = \frac{(x-2)(x+4)}{(x-6)(x-2)}$	2. (a) Domain 4 pts (b) Hole 4 pts (b) Vertical 3 pts (b) Horizontal 3 pts (c) x-intercept 3 pts (c) y-intercept 3 pts
3. Find all zeros: (a) list Rational Root Theorem candidates, then test only -1 and 1; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros. $p(x) = x^3 + 4x^2 + 4x + 1$	3. (a) Candidates/root 5 pts (b) Quotient 7 pts (c) Other zeros 8 pts

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QUESTIONS	ANSWERS
4. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain. $f(x) = 2\sqrt{x+4} - 6$	4. (a) Domain 4 pts (a) Range 4 pts (b) x-intercept 3 pts (b) y-intercept 3 pts (c) Infinite end 4 pts (c) Finite end 2 pts
5. Write the least-degree polynomial in factored form matching the graph and labeled point. State its end behavior in both directions. 	5. Polynomial 16 pts Left end 2 pts Right end 2 pts

Instructor answer key: MATH 101 Unit 3: Polynomial, Rational, and Radical Functions

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. (a) Vertex
$(0,\ -18)$ — 6 pts

(a) Minimum
-18 — 4 pts

(b) Zeros
$x=-3,\ 3$ — 10 pts

2. (a) Domain
$\mathbb{R}\setminus\{2,6\}$ — 4 pts

(b) Hole
$(2,\ -\frac{3}{2})$ — 4 pts

(b) Vertical
$x=6$ — 3 pts

(b) Horizontal
$y=1$ — 3 pts

(c) x-intercept
$(-4,\ 0)$ — 3 pts

(c) y-intercept
$(0,\ -\frac{2}{3})$ — 3 pts

3. (a) Candidates/root
$\{-1,1\};\ x=-1$ — 5 pts

(b) Quotient
$x^2+3x+1;\ R=0$ — 7 pts

(c) Other zeros
$x=-\frac{3}{2}-\frac{\sqrt{5}}{2},\ -\frac{3}{2}+\frac{\sqrt{5}}{2}$ — 8 pts

TEST PAGE 2

4. (a) Domain
$[-4,\infty)$ — 4 pts

(a) Range
$[-6,\infty)$ — 4 pts

(b) x-intercept
$(5,\ 0)$ — 3 pts

(b) y-intercept
$(0,\ -2)$ — 3 pts

(c) Infinite end
$x\to\infty:\ f(x)\to\infty$ — 4 pts

(c) Finite end
$x\to -4^{+}:\ f(x)\to -6$ — 2 pts

5. Polynomial
$-1(x+1)(x-2)^2$ — 16 pts

Left end
$x\to -\infty:\ f(x)\to\infty$ — 2 pts

Right end
$x\to\infty:\ f(x)\to -\infty$ — 2 pts